2.1) Suppose

$$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$$

Compute

$$x^T \Sigma^{-1} x$$

for $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$$\Sigma^{-1} = \begin{bmatrix} 1/4 & 0 \\ 0 & 1 \end{bmatrix}$$

$$x^T \Sigma^{-1} x = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 1/4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 \end{bmatrix} \left(2 \begin{bmatrix} 1/4 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$$

$$= 1 + 1 = 2$$

8.1) Given

$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Sigma = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix}$

Compute the squared Mahalanobis distance of

$x = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ $\qquad d^2 = (x-\mu)^T \Sigma^{-1} (x-\mu)$

$\Sigma^{-1} = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 1/9 & 0 \\ 0 & 1/4 \end{bmatrix}$

$d^2 = \begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} 1/9 & 0 \\ 0 & 1/4 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

$d^2 = \begin{bmatrix} 3 & 2 \end{bmatrix} \left(3 \begin{bmatrix} 1/9 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1/4 \end{bmatrix} \right)$

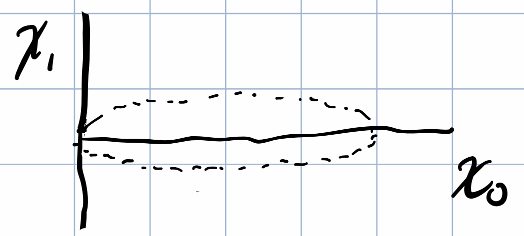
$d^2 = \begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/2 \end{bmatrix}$

$d^2 = 1 + 1 = 2$

8.2) Without doing any computation, sketch the covariance elipse for

$$\Sigma = \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix}.$$

Which axis is longer? Approximately how long are each semi-axis?



$$\frac{x_0}{x_1} = 4$$ x_0 is longer

8.3) Suppose an eigen value becomes very small (approaching zero). What happens to

a) the corresponding axis of the covariance ellipsoid

it shrinks to zero, with the over-all effect of going from an ellipse to a line

b) the corresponding entry in the precision matrix

it approaches infinity

c) the Mahalanobis penalty for moving in that direction?

Moving in that direction would blow up, making this an extremely unexpected event.